

LAYER: 3_MesMDM

This file is a merged representation of a subset of the codebase, containing files not matching ignore patterns, combined into a single document by Repomix.

<file_summary>

This section contains a summary of this file.

<purpose>

This file contains a packed representation of a subset of the repository's contents that is considered the most important context.

It is designed to be easily consumable by AI systems for analysis, code review, or other automated processes.

</purpose>

<file_format>

The content is organized as follows:

1. This summary section
2. Repository information
3. Directory structure
4. Repository files (if enabled)
5. Multiple file entries, each consisting of:
 - File path as an attribute
 - Full contents of the file

</file_format>

<usage_guidelines>

- This file should be treated as read-only. Any changes should be made to the original repository files, not this packed version.
- When processing this file, use the file path to distinguish between different files in the repository.
- Be aware that this file may contain sensitive information. Handle it with the same level of security as you would the original repository.

</usage_guidelines>

<notes>

- Some files may have been excluded based on .gitignore rules and Repomix's configuration
- Binary files are not included in this packed representation. Please refer to the Repository Structure section for a complete list of file paths, including binary files
- Files matching these patterns are excluded: **/obj/**, **/bin/**, **/*.png, **/*.jpg, **/node_modules/**, **/*.Designer.cs
- Files matching patterns in .gitignore are excluded
- Files matching default ignore patterns are excluded
- Files are sorted by Git change count (files with more changes are at the bottom)

</notes>

</file_summary>

<directory_structure>

Promatis.Net.MES.MDM.Configuration/Class1.cs
Promatis.Net.MES.MDM.Configuration/Promatis.Net.MES.MDM.Configuration.csproj
Promatis.Net.MES.MDM.Data/DBContext/MesMdmApplicationDbContext.cs
Promatis.Net.MES.MDM.Data/Promatis.Net.MES.MDM.Data.csproj
Promatis.Net.MES.MDM.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs
Promatis.Net.MES.MDM.Domain.Interface/Class1.cs
Promatis.Net.MES.MDM.Domain.Interface/Promatis.Net.MES.MDM.Domain.Interface.csproj
Promatis.Net.MES.MDM.Domain/Promatis.Net.MES.MDM.Domain.csproj
Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperation.cs
Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperationValidator.cs
Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/TechnologicalOperationParameter.cs
Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameterValidator.cs
Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnit.cs
Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnitValidator.cs
Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameter.cs
Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameterValidator.cs
Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/TechnologicalParameterCalcMethod.cs
Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodValidator.cs
Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBase.cs
Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBaseValidator.cs
Promatis.Net.MES.MDM.Domain/Units/PositionUnitBase.cs
Promatis.Net.MES.MDM.Domain/Units/PositionUnitBaseValidator.cs
Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBase.cs
Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBaseValidator.cs
Promatis.Net.MES.MDM.Domain/Units/StorageUnitBase.cs

```
Promatis.Net.MES.MDM.Domain/Units/StorageUnitBaseValidator.cs
Promatis.Net.MES.MDM.Domain/Units/TransportUnitBase.cs
Promatis.Net.MES.MDM.Domain/Units/TransportUnitBaseValidator.cs
Promatis.Net.MES.MDM.Service/Promatis.Net.MES.MDM.Service.csproj
Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationParameterService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterCalcMethodService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterService.cs
Promatis.Net.MES.MDM.Tests/Promatis.Net.MES.MDM.Tests.csproj
Promatis.Net.MES.MDM.Tests/UnitCategoryValidationTests.cs
Promatis.Net.MES.MDM.Tests/UnitTest1.cs
Promatis.Net.MES.MDM.UI/_Imports.razor
Promatis.Net.MES.MDM.UI/Promatis.Net.MES.MDM.UI.csproj
Promatis.Net.MES.MDM.UI/UiModule.cs
Promatis.Net.MES.MDM.UI/wwwroot/exampleJsInterop.js
</directory_structure>
```

<files>

This section contains the contents of the repository's files.

<file path="Promatis.Net.MES.MDM.Configuration/Class1.cs">

```
namespace Promatis.Net.MES.MDM.Configuration
{
    public class Class1
    {
    }
}
```

<file path="Promatis.Net.MES.MDM.Configuration/Promatis.Net.MES.MDM.Configuration.csproj">

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <TargetFramework>net10.0</TargetFramework>
    <ImplicitUsings>enable</ImplicitUsings>
    <Nullable>enable</Nullable>
  </PropertyGroup>
  <ItemGroup>
    <ProjectReference Include="..\..\..\
\MesCore\Promatis.Net.MES.Configuration\Promatis.Net.MES.Configuration.csproj" />
    <ProjectReference Include="..
\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />
  </ItemGroup>
</Project>
</file>
```

<file path="Promatis.Net.MES.MDM.Data/DBContext/MesMdmApplicationDbContext.cs">

```
using Microsoft.EntityFrameworkCore;
using Microsoft.Extensions.Configuration;
using Promatis.Net.MES.Data;
using Promatis.Net.MES.MDM.Domain;
namespace Promatis.Net.MES.MDM.Data;
public class MesMdmApplicationDbContext(
    DbContextOptions options,
    IConfiguration configuration,
    IServiceProvider? serviceProvider = null) : MesApplicationDbContext(options, configuration, serviceProvider)
{
    public DbSet<TechnologicalOperation> TechnologicalOperations => Set<TechnologicalOperation>();
    public DbSet<TechnologicalOperationUnit> TechnologicalOperationUnits =>
    Set<TechnologicalOperationUnit>();
    public DbSet<TechnologicalParameter> TechnologicalParameters => Set<TechnologicalParameter>();
    public DbSet<TechnologicalOperationParameter> TechnologicalOperationParameters =>
    Set<TechnologicalOperationParameter>();
    public DbSet<TechnologicalParameterCalcMethod> TechnologicalParameterCalcMethods =>
    Set<TechnologicalParameterCalcMethod>();
}
```

<file path="Promatis.Net.MES.MDM.Data/Promatis.Net.MES.MDM.Data.csproj">

```
<Project Sdk="Microsoft.NET.Sdk">
```

```

    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>
    <ItemGroup>
        <PackageReference Include="Microsoft.EntityFrameworkCore" Version="10.0.8" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.Relational" Version="10.0.8" />
        <PackageReference Include="Microsoft.Extensions.Configuration" Version="10.0.8" />
    </ItemGroup>
    <ItemGroup>
        <ProjectReference Include="..\..\..\
\MesCore\Promatis.Net.MES.Data\Promatis.Net.MES.Data.csproj" />
        <ProjectReference Include="..\Promatis.Net.MES.MDM.Domain\Promatis.Net.MES.MDM.Domain.csproj" />
    </ItemGroup>
</Project>
</file>

<file path="Promatis.Net.MES.MDM.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs">
using Promatis.Net.MES.Data;
using Promatis.Net.MES.MDM.Domain;
namespace Promatis.Net.MES.MDM.Data;
/// <summary>
/// A=C05Dt=Cä5 C0@C,,:C´0CD=Cä5 D 5D,,5C08CR 4C´0 D$@C,,3C45D 0 C,,5D 0D EC,,8 D$5DT=Cä;Cä3C,,GCTAC=8DR >C05D 0D
/// </summary>
public class TechnologicalOperationHierarchyTrigger : TechnologicalOperationHierarchyTrigger<
    TechnologicalOperation,
    TechnologicalOperationUnit,
    TechnologicalOperationParameter>
{
}
</file>

<file path="Promatis.Net.MES.MDM.Domain.Interface/Class1.cs">
namespace Promatis.Net.MES.MDM.Domain.Interface
{
    public class Class1
    {
    }
}
</file>

<file path="Promatis.Net.MES.MDM.Domain.Interface/Promatis.Net.MES.MDM.Domain.Interface.csproj">
<Project Sdk="Microsoft.NET.Sdk">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>
    <ItemGroup>
        <ProjectReference Include="..\..\..\
\MesCore\Promatis.Net.MES.Domain.Interface\Promatis.Net.MES.Domain.Interface.csproj" />
    </ItemGroup>
</Project>
</file>

<file path="Promatis.Net.MES.MDM.Domain/Promatis.Net.MES.MDM.Domain.csproj">
<Project Sdk="Microsoft.NET.Sdk">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>
    <ItemGroup>

```

```

        <ProjectReference Include="..\..\..\
\MesCore\Promatis.Net.MES.Domain\Promatis.Net.MES.Domain.csproj" />
        <ProjectReference Include="..\
\Promatis.Net.MES.MDM.Domain.Interface\Promatis.Net.MES.MDM.Domain.Interface.csproj" />
    </ItemGroup>
}
</Project>
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperation.cs">
using Promatis.Net.MES.Domain;
}
namespace Promatis.Net.MES.MDM.Domain;
}
public class TechnologicalOperation : TechnologicalOperationBase<
    TechnologicalOperation,
    TechnologicalOperationUnit,
    TechnologicalOperationParameter>
{
    // A;C AD 0C ACä;DäBCô> Dt8D B. }
    // Aä= C 2D$>CÄ0D$8Dt5D :C, ACä4CT@Cd8D" AC$>C"AD$2C &VçB 8 Children D BC,,?Cä< TechnologicalOperation
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperationValidator.cs">
using Promatis.Net.MES.Domain;
}
namespace Promatis.Net.MES.MDM.Domain;
}
public class TechnologicalOperationValidator :
    TechnologicalOperationBaseValidator<TechnologicalOperation>
{
    }
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameter.cs">
using Promatis.Net.MES.Domain;
}
namespace Promatis.Net.MES.MDM.Domain;
}
public class TechnologicalOperationParameter :
    TechnologicalOperationParameterBase<TechnologicalOperation, TechnologicalParameter> { }
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameterValidator.cs">
using Promatis.Net.MES.Domain;
}
namespace Promatis.Net.MES.MDM.Domain;
}
public class TechnologicalOperationParameterValidator :
    TechnologicalOperationParameterBaseValidator<TechnologicalOperationParameter,
    TechnologicalOperation, TechnologicalParameter>
{
    }
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnit.cs">
using Promatis.Net.MES.Domain;
}
namespace Promatis.Net.MES.MDM.Domain;
}
public class TechnologicalOperationUnit : TechnologicalOperationUnitBase<TechnologicalOperation>
{
    /// <summary>
    /// Aô5D 5C@D´2C 5CÄ 1C 7Cä2Cä5 D 2Cä9D BC$> B #A D BD >C4> D$8Cö8Ct8D >C$0C0=D´< D 2Cä9D BC$>CÄ 4C´0
    /// </summary>
    public new virtual TechnologicalOperation Operation
    {
        get => base.Operation as TechnologicalOperation ?? null!;
        set => base.Operation = value;
    }
}

```

</file>

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/
TechnologicalOperationUnitValidator.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class TechnologicalOperationUnitValidator :
TechnologicalOperationUnitBaseValidator<TechnologicalOperationUnit, TechnologicalOperation>
{
    {
}
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameter.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class TechnologicalParameter : TechnologicalParameterBase
{
    // Bt8D BD'9 C;C AD 4C'O DbSet<TechnologicalParameter>
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameterValidator.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class TechnologicalParameterValidator :
TechnologicalParameterBaseValidator<TechnologicalParameter>
{
    {
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethod.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
/// <summary>
/// A>C0:D 5D$=C 0 D 5C ;C,,7C FC,,0 CÄ5D$>CD0 D 0D GCTBC BCTEC0>C'>C48Dt5D :Cä3Cä ?C @C <CTBD 0.D
/// B 2D67D'2C 5D" :Cä=C@CTBC0CDâ >C05D 0Dd8DâÂ >C >D CCD>C$0C08CR 8 C00D 0CÄ5D$@ C,,7 C@>C0BCT:D BC ÖW40D0i
/// </summary>
public class TechnologicalParameterCalcMethod : TechnologicalParameterCalcMethodBase<UnitBase,
TechnologicalOperation, TechnologicalParameter>
{
    // A;C AD GC,,AD" 8 C4>D$>C" : CÄ0C0?C,,=C4C C" F$6öçFW†B <Cä4D4;D0 ÖW40D0i
    // B 2Cä9D BC$0 Unit, TechnologicalOperation C, FV6†æ0Æöv-6 Ä & ÖWFW"
    // C 2D$>CÄ0D$8Dt5D :C, ?Cä;D4GC,,;C, AD$@Cä3C,,5 C@>C05Dt=D'5 D$8C0K.
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodValidator.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class TechnologicalParameterCalcMethodValidator :
TechnologicalParameterCalcMethodBaseValidator<
    TechnologicalParameterCalcMethod,
    UnitBase,
    TechnologicalOperation,
    TechnologicalParameter>
{
    {
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBase.cs">
using Promatis.Net.MES.Domain;
using Promatis.Net.MES.Domain.Interface;
{
}
```

```

Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public abstract class DepartmentUnitBase : UnitBaseÐ
{Ð
    protected DepartmentUnitBase() => Kind = UnitKind.Department;Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public abstract class DepartmentUnitBaseValidator<T> : UnitBaseValidator<T> where T :
DepartmentUnitBaseÐ
{Ð
}
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/PositionUnitBase.cs">
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public abstract class PositionUnitBase : UnitBaseÐ
{Ð
    protected PositionUnitBase() => Kind = UnitKind.Position;Ð
}
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/PositionUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class PositionUnitBaseValidator<T> : UnitBaseValidator<T> where T : PositionUnitBaseÐ
{Ð
}
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBase.cs">
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public abstract class ProductionUnitBase : UnitBaseÐ
{Ð
    protected ProductionUnitBase() => Kind = UnitKind.Production;Ð
}
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class ProductionUnitBaseValidator<T> : UnitBaseValidator<T> where T : ProductionUnitBaseÐ
{Ð
}
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/StorageUnitBase.cs">
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public abstract class StorageUnitBase : UnitBaseÐ
{Ð
    protected StorageUnitBase() => Kind = UnitKind.Storage;Ð
}
}
</file>

```

```
<file path="Promatis.Net.MES.MDM.Domain/Units/StorageUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class StorageUnitBaseValidator<T> : UnitBaseValidator<T> where T : StorageUnitBase
{
}
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/Units/TransportUnitBase.cs">
using Promatis.Net.MES.Domain;
using Promatis.Net.MES.Domain.Interface;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public abstract class TransportUnitBase : UnitBase
{
protected TransportUnitBase() => Kind = UnitKind.Transport;
}
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/Units/TransportUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;
{
namespace Promatis.Net.MES.MDM.Domain;
{
public class TransportUnitBaseValidator<T> : UnitBaseValidator<T> where T : TransportUnitBase
{
}
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Service/Promatis.Net.MES.MDM.Service.csproj">
<Project Sdk="Microsoft.NET.Sdk">
{
<PropertyGroup>
<TargetFramework>net10.0</TargetFramework>
<ImplicitUsings>enable</ImplicitUsings>
<Nullable>enable</Nullable>
</PropertyGroup>
{
<ItemGroup>
<ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.Service\Promatis.Net.MES.Service.csproj" />
<ProjectReference Include="..\Promatis.Net.MES.MDM.Data\Promatis.Net.MES.MDM.Data.csproj" />
</ItemGroup>
}
</Project>
</file>
```

```
<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationParameterService.cs">
using Microsoft.EntityFrameworkCore;
using Promatis.Net.MES.MDM.Data;
using Promatis.Net.MES.MDM.Domain;
using Promatis.Net.MES.Service;
{
namespace Promatis.Net.MES.MDM.Service;
{
/// <summary>
/// BD8C00C'LC0KC' ACT@C$8D 4C'O D4?D 0C$;CT=C,,O C0@C,,2D67C=0CÄ8 C00D 0CÄ5D$@Cä2 C$ >C05D 0Dd8D0<.D
/// </summary>
public class TechnologicalOperationParameterService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)
: TechnologicalOperationParameterService<
TechnologicalOperationParameter,
TechnologicalOperation,
TechnologicalParameter,
MesMdmApplicationDbContext>(contextFactory)
{
// A=C AD 3CäBCä2 C$ 8D ?Cä;DÄ7Cä2C =C,,N C" 1C,,7C05D 0;Cä3C,,:CR 8 UI.D
}
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationService.cs">
using Microsoft.EntityFrameworkCore;
using Promatis.Net.MES.MDM.Data;
{
namespace Promatis.Net.MES.MDM.Service;
{
public class TechnologicalOperationService
{
}
}
</file>
```

```

using Promatis.Net.MES.MDM.Domain;
using Promatis.Net.MES.Service;
{
namespace Promatis.Net.MES.MDM.Service;
{
/// <summary>
/// A=C05Dt=C 0 C@C,,:C'0CD=C 0 D ;D46C 0 CD;D0 CC0@C 2C'5C08D0 BCTEC0>C'>C48Dt5D :C,,<C, >C05D 0Dd8D0<C,i
/// A,,!A0 A A' A0 : At0C@D'2C 5D" BD >C'=Cä9 CD6CT=CT@C,,: C 1D BD 0C=BC0>C4> D 5D 2C,,AC 0CD@C @CT0C'LC0KC
/// </summary>
public class TechnologicalOperationService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)
{
: TechnologicalOperationService<TechnologicalOperation, TechnologicalOperationUnit,
MesMdmApplicationDbContext>(contextFactory)
{
/// <summary>
/// A0@C,,:C'0CD=C 0 DD0C @C,,:C ?Cä;C,,<Cä@DD8Ct<C â
/// A=C0AD$@D48D CCTB Dt8D BD'9 CD>Dt5D =C,,9 D,,0C ;Cä= C0> C$ACT< C0@C 2C,,;C < C# C, &W V-&VB0AC$>C"AD$2
/// </summary>
public override Task<TechnologicalOperation> CreateChildTemplateAsync(TechnologicalOperation
parent)
{
if (parent == null) throw new ArgumentNullException(nameof(parent));

// At0D"8D$0 CD>CÄ5C00: CTAC'8 D >CD8D$5C'L D 0CÄ 0C$;D05D$AD0 ;C,,AD$>CÄÄ 3CT=CT@C,,@D45CÄ 8D :C'NDt5C
if (parent.IsLeaf)
{
throw new InvalidOperationException(
$"A05C$>Ct<Cä6C0> D >Ct4C BDÄ 4CäGCT@C08C' MC'5CÄ5C0B. Aä?CT@C FC,,0 '{parent.Name}' D02C'OCTB
)
}

// A,,=C,,FC,,0C'8Ct8D CCT< required-D 2Cä9D BC$0 D >C4;C AC0> D ?CTFC,,DC,,:C FC,,8 C=CÄ?C,,;D0BCä@C0>C4>
var childOperation = new TechnologicalOperation
{
Id = Guid.NewGuid(),
ParentId = parent.Id,
Parent = null, // B 0Ct@D'2C 5CÄ AD KC':D2 2 Aä B2 4C'O Dt8D BCä9 UI-D 5D AC,,8D

// A0> D4<Cä;Dt0C08Dä 2C0CD$@C, 3D CC0?D2 ACä7CD0CT< C=C05Dt=D4N C BCä<C @C0CDä >C05D 0Dd8Dä ,, C
IsLeaf = true,

Code = string.Empty,
Name = string.Empty,
Description = string.Empty
};

return Task.FromResult(childOperation);
}
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterCalcMethodService.cs">
using Microsoft.EntityFrameworkCore;
using Promatis.Net.MES.Domain;
using Promatis.Net.MES.MDM.Data;
using Promatis.Net.MES.MDM.Domain;
using Promatis.Net.MES.Service;
{
namespace Promatis.Net.MES.MDM.Service;
{
/// <summary>
/// BD8C00C'LC0KC' ACT@C$8D 4C'O D4?D 0C$;CT=C,,0 C,,=D BD CC=FC,,0CÄ8 C, <CTBCä4C <C, @C Adt5D$0 D$5C'5CÄ5D$@C
/// </summary>
public class TechnologicalParameterCalcMethodService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)
{
: TechnologicalParameterCalcMethodService<
TechnologicalParameterCalcMethod,
UnitBase,
TechnologicalOperation,
TechnologicalParameter,
MesMdmApplicationDbContext>(contextFactory)
{
// A0>C'=D'9 D ?CT:D$@ C 8Ct=CTA-C'>C48C=8 CD;D0 =C AD$@Cä9C=8 C 3D 5C40Dd8C, 4C =C0KDR 3CäBCä2.D
}
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterService.cs">

```



```

using Microsoft.EntityFrameworkCore;
using Promatis.Net.MES.Data;
using Promatis.Net.MES.MDM.Data;
using Promatis.Net.MES.MDM.Domain;
using Promatis.Net.MES.Service;

namespace Promatis.Net.MES.MDM.Service;

/// <summary>
/// BD8C00C'LC0KC' ACT@C$8D 4C'O D 0C >D$K D > D ?D 0C$>Dt=C,,:Cä< D$5DT=Cä;Cä3C,,GCTACª8DR ?C @C <CTBD >C" 2
/// </summary>
public class TechnologicalParameterService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)
    : TechnologicalParameterService<TechnologicalParameter,
MesMdmApplicationDbContext>(contextFactory)
{
    public override async Task<List<TechnologicalParameter>> GetAllAsync()
    {
        await using MesApplicationDbContext context = await ContextFactory.CreateDbContextAsync();

        return await context.Set<TechnologicalParameter>()
            .Include(x => x.UnitOfMeasurement) // Ad0CD=C 0 Ct0C4@D47Cª0 D ?D 0C$>Dt=C,,:C CBâ Ct<.D
            .ToListAsync();
    }
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Tests/Promatis.Net.MES.MDM.Tests.csproj">
<Project Sdk="Microsoft.NET.Sdk">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
        <IsPackable>false</IsPackable>
    </PropertyGroup>

    <ItemGroup>
        <PackageReference Include="coverlet.collector" Version="10.0.0">
            <PrivateAssets>all</PrivateAssets>
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>
        </PackageReference>
        <PackageReference Include="FluentAssertions" Version="8.10.0" />
        <PackageReference Include="FluentValidation" Version="12.1.1" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.InMemory" Version="10.0.8" />
        <PackageReference Include="Microsoft.NET.Test.Sdk" Version="18.5.1" />
        <PackageReference Include="Moq" Version="4.20.72" />
        <PackageReference Include="xunit.runner.visualstudio" Version="3.1.5">
            <PrivateAssets>all</PrivateAssets>
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>
        </PackageReference>
        <PackageReference Include="xunit.v3" Version="3.2.2" />
    </ItemGroup>

    <ItemGroup>
        <ProjectReference Include="..
\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />
    </ItemGroup>

    <ItemGroup>
        <Using Include="Xunit" />
    </ItemGroup>
</Project>
</file>

```

```

<file path="Promatis.Net.MES.MDM.Tests/UnitCategoryValidationTests.cs">
using FluentAssertions;
using FluentValidation.TestHelper;
using Promatis.Net.MES.Domain.Interface;
using Promatis.Net.MES.MDM.Domain;

namespace Promatis.Net.MES.MDM.Tests;

```

```

D
// B 5C ;C,,7C FC,,8 D CD"=CäAD$5C•
public class TestDept : DepartmentUnitBase { }D
public class TestProd : ProductionUnitBase { }D
public class TestStor : StorageUnitBase { }D
public class TestPos : PositionUnitBase { }D
public class TestTrans : TransportUnitBase { }D
D
// B 5C ;C,,7C FC,,8 C$0C`8CD0D$>D >C-
public class TestDeptValidator : DepartmentUnitBaseValidator<TestDept> { }D
public class TestProdValidator : ProductionUnitBaseValidator<TestProd> { }D
public class TestStorValidator : StorageUnitBaseValidator<TestStor> { }D
public class TestPosValidator : PositionUnitBaseValidator<TestPos> { }D
public class TestTransValidator : TransportUnitBaseValidator<TestTrans> { }D
D
public class UnitCategoryValidationTestsD
{D
    private readonly TestDeptValidator _deptValidator = new();D
    private readonly TestProdValidator _prodValidator = new();D
    private readonly TestStorValidator _storValidator = new();D
    private readonly TestPosValidator _posValidator = new();D
    private readonly TestTransValidator _transValidator = new();D
D
    [Theory]D
    [InlineData(UnitType.Workshop, true)] // Aä: CD;Dò CT?C @D$0CÄ5C0BC
    [InlineData(UnitType.Section, true)] // Aä: CD;Dò CT?C @D$0CÄ5C0BC
    [InlineData(UnitType.Crane, false)] // AäHC,,1C@0: A@C = C05 CÄ>Cd5D" 1D`BDÂ CT?C @D$0CÄ5C0BCä<D
    public void DepartmentValidator_Should_Verify_Types(UnitType type, bool isValid)D
    {D
        var unit = new TestDept { Type = type, Name = "Dept" };D
        TestValidationResult<TestDept>? result = _deptValidator.TestValidate(unit);D
D
        if (isValid) result.ShouldNotHaveAnyValidationErrors();D
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C@0D
    }D
D
    [Theory]D
    [InlineData(UnitType.MachineTool, true)] // Aä: CD;Dò &öGV7F-öí
    [InlineData(UnitType.Table, true)] // Aä: CD;Dò &öGV7F-öí
    [InlineData(UnitType.Storage, false)] // AäHC,,1C@0: B :C`0CB =CR <Cä6CTB C KD$L ProductionD
    public void ProductionValidator_Should_Verify_Types(UnitType type, bool isValid)D
    {D
        var unit = new TestProd { Type = type, Name = "Prod" };D
        TestValidationResult<TestProd>? result = _prodValidator.TestValidate(unit);D
D
        if (isValid) result.ShouldNotHaveAnyValidationErrors();D
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C@0D
    }D
D
    [Theory]D
    [InlineData(UnitType.Cell, true)] // BôGCT9C@0 - DôBCâ C 4C`O B :C`0CD0D
    [InlineData(UnitType.Rack, true)] // B BCT;C`0Cb r MD$> Aä: CD;Dò !C@;C 4C
    [InlineData(UnitType.Vehicle, false)] // B$@C =D ?Cä@D" r =CR !C@;C 4D
    public void StorageValidator_Should_Verify_Types(UnitType type, bool isValid)D
    {D
        var unit = new TestStor { Type = type, Name = "Stor" };D
        TestValidationResult<TestStor>? result = _storValidator.TestValidate(unit);D
D
        if (isValid) result.ShouldNotHaveAnyValidationErrors();D
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C@0D
    }D
D
    [Theory]D
    [InlineData(UnitType.Cell, true)] // BôGCT9C@0 CÄ>Cd5D" 1D`BDÂ Cä7C,,FC,,5C•
    [InlineData(UnitType.Other, true)] // Other CÄ>Cd5D" 1D`BDÂ Cä7C,,FC,,5C•
    [InlineData(UnitType.Workshop, false)] // Bd5DR =CR <Cä6CTB C KD$L Aô>Ct8Dd8CT9D
    public void PositionValidator_Should_Verify_Types(UnitType type, bool isValid)D
    {D
        var unit = new TestPos { Type = type, Name = "Pos" };D
        TestValidationResult<TestPos>? result = _posValidator.TestValidate(unit);D
D
        if (isValid) result.ShouldNotHaveAnyValidationErrors();D
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C@0D
    }D
D
    [Theory]D

```

```

[InlineData(UnitType.Vehicle, true)] // Aö>C4@D47Dt8C¢ r C¢ 4C´O B$@C =D ?Cä@D$0Ð
[InlineData(UnitType.Conveyor, true)] // Aα>CÔ2CT9CT@ – Aä: CD;Dò "D 0CÔACô>D BC
[InlineData(UnitType.MachineTool, false)] // B BC =Cä: – CÔ5 B$@C =D ?Cä@D-
[InlineData(UnitType.Cell, false)] // BôGCT9Cα0 – CÔ5 B$@C =D ?Cä@D-
public void TransportValidator_Should_Verify_Types(UnitType type, bool isValid)Ð
{Ð
    var unit = new TestTrans { Type = type, Name = "Trans" };Ð
    TestValidationResult<TestTrans>? result = _transValidator.TestValidate(unit);Ð
Ð
    if (isValid) result.ShouldNotHaveAnyValidationErrors();Ð
    else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB Cα0D
}Ð
Ð
Ð
[Fact]Ð
public void Kind_Should_Be_Automatically_Set_By_Base_Class()Ð
{Ð
    // Aô@Cä2CT@Dô5CÂÂ GD$> CÔ0D,,8 CÔ>C$KCR 0C AD$@C :D$=D´5 Cα;C AD K D 0CÄ8 Cô@CäAD$0C$;DôND" ¶-æBí
    // Aô5D 5CD0CT< Cä1Dô7C BCT;DÄ=D´9 Type C, æ ÖR 4C´O D >C ;Dä4CT=C,,0 Cä3D 0CÔ8Dt5CÔ8C´ &W V-&VBí
Ð
    new TestDept { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Department);Ð
    new TestProd { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Production);Ð
    new TestStor { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Storage);Ð
    new TestPos { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Position);Ð
    new TestTrans { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Transport);Ð
}Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Tests/UnitTest1.cs">
namespace Promatis.Net.MES.MDM.TestsÐ
{Ð
    public class UnitTest1Ð
    {Ð
        [Fact]Ð
        public void Test1()Ð
        {Ð
Ð
        }Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.MDM.UI/_Imports.razor">
@using Microsoft.AspNetCore.Components.FormsÐ
@using Microsoft.AspNetCore.Components.RoutingÐ
@using Microsoft.AspNetCore.Components.WebÐ
@using MudBlazorÐ
@using Promatis.Net.UI.Components.WorkspacesÐ
@using Promatis.Net.UI.Components.Dialogs
</file>

<file path="Promatis.Net.MES.MDM.UI/Promatis.Net.MES.MDM.UI.csproj">
<Project Sdk="Microsoft.NET.Sdk.Razor">Ð
Ð
    <PropertyGroup>Ð
        <TargetFramework>net10.0</TargetFramework>Ð
        <Nullable>enable</Nullable>Ð
        <ImplicitUsings>enable</ImplicitUsings>Ð
    </PropertyGroup>Ð
Ð
    <ItemGroup>Ð
        <Compile Remove="Pages\TechnologicalOperations\*" />Ð
        <Content Remove="Pages\TechnologicalOperations\*" />Ð
        <EmbeddedResource Remove="Pages\TechnologicalOperations\*" />Ð
        <None Remove="Pages\TechnologicalOperations\*" />Ð
    </ItemGroup>Ð
Ð
Ð
    <ItemGroup>Ð
        <SupportedPlatform Include="browser" />Ð
    </ItemGroup>Ð
Ð
    <ItemGroup>Ð
        <PackageReference Include="Microsoft.AspNetCore.Components.Web" Version="10.0.8" />Ð
        <PackageReference Include="MudBlazor" Version="9.4.0" />Ð
    </ItemGroup>

```

```

    </ItemGroup>
  }
  <ItemGroup>
    <ProjectReference Include="..\..\..\
\Core\Configuration.Web\Promatis.Net.Configuration.Web.csproj" />
    <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.UI\Promatis.Net.MES.UI.csproj" />
    <ProjectReference Include="..\
\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />
  </ItemGroup>
}
<ItemGroup>
  <Folder Include="Pages\TechnologicalParameter\Card\" />
</ItemGroup>
}
</Project>
</file>

<file path="Promatis.Net.MES.MDM.UI/UiModule.cs">
using MudBlazor;
using Promatis.Net.UI;
{
namespace Promatis.Net.MES.MDM.UI;
{
public class UiModule : IUiModule
{
    public string Name => GetType().Assembly.GetName().Name ?? "Unknown.Module";

    public IEnumerable<(string Title, string Href, string Icon, string? Group)> GetMenuItems()
    {
        return new List<(string Title, string Href, string Icon, string? Group)>
        {
            // B ?D 0C$>Dt=C,,: D$5DT=Cä;Cä3C,,GCTAC=8DR >Cö5D 0Dd8C•
            ("B$5DT=Cä;Cä3C,,GCTAC=8CR >Cö5D 0Dd8C,"Ä "öÖW2öÖFö÷FV6†æöÆöv-6 ÄÖ÷ W& F-öç2"Ä
Icons.Material.Filled.SettingsInputComponent, "B ?D 0C$>Dt=C,,:C,"Í
        {
            // Aö;CäAC=8C' ACô@C 2CäGCö8C¢ BCTECô>C´>C48Dt5D :C,,E Cö0D 0CÄ5D$@Cä2ö
            ("B$5DT=Cä;Cä3C,,GCTAC=8CR ?C @C <CTBD K", "/mes/mdm/technological-parameters",
Icons.Material.Filled.SettingsInputComponent, "B ?D 0C$>Dt=C,,:C,"•
        }
    }
}
}
</file>

<file path="Promatis.Net.MES.MDM.UI/wwwroot/exampleJsInterop.js">
// This is a JavaScript module that is loaded on demand. It can export any number of
// functions, and may import other JavaScript modules if required.
{
export function showPrompt(message) {
    return prompt(message, 'Type anything here');
}
}
</file>

</files>

```